

TAIYUAN, China

WMO: 537720

Lat: 37.6206N

Lon: 112.5764E

Elev: 2575

StdP: 13.38

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	10.0	-16.6	2.4	21.6	-12.7	3.0	23.3	21.8	29.1	18.8	28.7	2.9	110	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	93.0	67.5	89.9	67.3	87.8	67.0	75.5	84.7	73.7	82.8	72.2	81.1	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	135.4	80.6	71.3	127.2	78.8	69.6	120.0	77.5	41.1	84.9	39.3	82.6	37.8	80.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	17.0	14.2	-0.7	97.5	3.6	2.9	-3.3	99.6	-5.4	101.3	-7.5	103.0	-10.1	105.1
			-2.0	78.3	3.4	2.2	-4.5	79.9	-6.5	81.2	-8.4	82.4	-10.9	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	24.5	31.6	43.2	55.7	66.2	73.3	76.8	73.8	64.7	52.9	39.0	27.4	(6)
(7)		DBStd	19.17	5.99	6.94	7.62	7.75	5.79	4.88	4.33	4.95	5.57	6.11	6.96	6.12	(7)
(8)		HDD50	2651	792	515	237	32	0	0	0	0	41	334	700		(8)
(9)		HDD65	5616	1257	935	677	290	56	5	0	4	70	376	781	1165	(9)
(10)		CDD50	3576	0	0	25	204	504	698	831	738	442	131	3	0	(10)
(11)		CDD65	1063	0	0	0	12	94	252	366	277	61	1	0	0	(11)
(12)		CDH74	10218	0	0	12	262	1242	2551	3331	2236	548	36	0	0	(12)
(13)		CDH80	3825	0	0	0	65	399	1066	1390	775	128	2	0	0	(13)
(14)	Wind	WSAvg	5.1	4.5	5.3	6.3	6.8	6.5	5.3	4.8	4.5	4.0	4.1	4.4	4.5	(14)
(15)	Precipitation	PrecAvg	17.6	0.1	0.2	0.5	0.9	1.3	2.2	4.3	3.9	2.4	1.0	0.6	0.1	(15)
(16)		PrecMax	24.6	0.6	0.7	2.1	2.4	3.0	4.4	9.5	8.3	4.7	2.8	2.0	0.9	(16)
(17)		PrecMin	10.2	0.0	0.0	0.0	0.0	0.2	0.3	1.4	0.4	0.3	0.2	0.0	0.0	(17)
(18)		PrecStd	3.4	0.2	0.2	0.6	0.6	0.7	1.0	2.0	2.1	1.3	0.7	0.6	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	59.0	75.1	86.9	91.8	96.6	96.6	93.0	88.7	78.4	64.8	49.4	(19)
(20)		0.4%	MCWB	34.8	41.4	50.4	58.3	60.3	65.5	69.9	70.8	66.1	58.0	47.4	36.1	(20)
(21)		2%	DB	43.8	53.2	68.2	80.6	87.9	92.3	93.2	89.6	83.8	73.5	60.6	45.2	(21)
(22)		2%	MCWB	31.9	38.8	46.9	54.1	58.6	63.9	70.1	70.6	63.8	55.2	44.9	33.6	(22)
(23)		5%	DB	39.9	49.1	63.0	76.5	84.4	89.5	90.0	87.4	80.3	69.7	56.4	42.0	(23)
(24)		5%	MCWB	29.8	36.2	44.5	53.2	58.0	63.7	70.4	70.3	62.7	53.9	43.1	31.8	(24)
(25)		10%	DB	37.0	44.9	58.8	71.8	80.9	86.2	87.7	84.4	76.8	65.8	51.9	38.9	(25)
(26)		10%	MCWB	28.1	33.7	42.6	51.3	56.8	63.4	70.4	69.4	61.8	52.1	41.1	30.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	43.2	52.7	62.3	65.6	72.5	78.2	77.3	72.0	61.6	50.2	37.7	(27)
(28)		0.4%	MCDWB	46.8	54.5	68.9	76.5	80.0	84.1	87.0	85.5	81.4	72.3	60.3	45.7	(28)
(29)		2%	WB	32.5	39.8	48.8	58.6	63.6	70.0	76.1	75.1	68.7	58.9	47.4	34.7	(29)
(30)		2%	MCDWB	42.4	50.5	64.3	74.0	77.8	82.0	85.6	84.1	76.5	67.4	55.9	42.8	(30)
(31)		5%	WB	30.4	37.2	46.1	55.8	61.7	68.2	74.6	73.4	65.8	56.8	44.9	32.6	(31)
(32)		5%	MCDWB	39.0	47.1	60.7	70.4	76.0	80.4	83.9	82.2	74.0	64.8	53.0	40.3	(32)
(33)		10%	WB	28.3	34.6	43.5	53.5	59.9	66.6	72.9	71.9	63.7	54.7	42.5	30.6	(33)
(34)		10%	MCDWB	36.0	44.1	56.8	67.1	74.8	78.9	82.0	80.3	72.6	62.4	50.9	37.8	(34)
(35)	Mean Daily Temperature Range		MDBR	23.1	23.6	24.9	26.0	25.8	23.9	20.1	19.6	21.8	23.6	22.5	21.2	(35)
(36)		5% DB	MCDBR	27.5	29.5	32.9	33.1	32.4	29.7	25.0	23.4	27.6	29.7	29.2	25.5	(36)
(37)		5% DB	MCWBR	18.2	18.0	17.6	14.3	12.3	10.2	9.1	9.4	11.2	14.1	16.4	16.2	(37)
(38)		5% WB	MCDBR	26.0	26.5	29.5	27.8	25.3	23.0	20.2	19.4	21.4	23.2	24.8	23.1	(38)
(39)		5% WB	MCWBR	17.5	16.4	16.5	13.5	11.4	9.9	9.6	9.5	10.8	12.5	14.8	15.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.405	0.491	0.512	0.551	0.589	0.689	0.679	0.661	0.599	0.545	0.431	0.388	(40)	
(41)		taud	1.997	1.812	1.799	1.757	1.729	1.586	1.652	1.667	1.743	1.799	1.981	2.064	(41)	
(42)		Ebn at Noon	232	224	236	236	230	207	208	207	210	205	220	228	(42)	
(43)		Edh at Noon	42	57	63	70	73	85	79	76	67	57	42	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	872	1070	1477	1769	1937	1779	1624	1522	1288	1116	906	794	(44)	
(45)		RadStd	79	114	132	128	136	139	113	119	109	117	121	70	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) -1.17	(i) -1.82	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page